PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence of Sarcopenia in Older Chinese Adults: A Systematic Review and Meta-analysis
AUTHORS	Xin, Caihong; Sun, Xin; Lu, Li; Shan, Ling

VERSION 1 – REVIEW

REVIEWER	Ming Yang Sichuan University, The Center of Gerontology and Geriatrics
REVIEW RETURNED	06-Aug-2020

GENERAL COMMENTS	Thank you for this invitation. This is a systematic review addressing the prevalence of sarcopenia in Chinese older adults. It has some merits; however, some important issues need to be addressed. 1. Most of the included studies were conducted in Chinese community-dwelling older adults. However, some of them were conducted in nursing homes or outpatient clinics. It is well-known that the prevalence of sarcopenia is significantly higher in nursing homes than in communities. Therefore, it is not a good idea to merge the data from different settings. Please consider only include the community-dwelling older adults or report the prevalence of sarcopenia in these settings separately. For the same reason, the expression of “the overall prevalence of sarcopenia among the elderly in China” in the abstract was inappropriate. 2. In the abstract, the authors stated that “the population-based studies that reported…” were selected. I fail to find a similar statement in the methods section. How did the authors know the study populations in the included studies were “population-based”? Generally, the studies conducted in outpatient clinics and nursing homes are not “population-based”. 3. The authors reported the search terms they used in the “search strategy” section. But this is not enough. They should also report the full search strategy for each database to improve the repeatability of the search process. Maybe a supplementary table would be added. Additionally, it is not enough to identify Chinese people using only “China” as the search term. 4. One inclusion criterion was “a cross-section study or retrospective study”. Were prospective studies excluded? At least, the baseline data of prospective studies are supposed to be valuable to the topic. 5. The full names of the databases -“VIP” and “Wanfang” should be given. 6. English writing needs to be improved. The expression of “the elderly” is a bit discriminatory, “older adults” would be better. Besides, the current running title “Prevalence of sarcopenia among” is inappropriate. 7. There are some typos. For example, “AWSG” and “IWSG” in Table 2.

REVIEWER	Min Cheol Chang Yeungnam University Medical Center
REVIEW RETURNED	25-Aug-2020

GENERAL COMMENTS	The authors evaluated the incidence of sarcopenia in China using meta-analysis. The idea of this study is novel and interesting. I think the manuscript is enough to be published. There is only one minor issue. In figure 2, the incidence of sarcopenia is 0.17 (mean), but in the body it is 0.18. The correction is needed.

REVIEWER	S Takahashi National Hospital Organisation Hamada Medical Center, Surgery
REVIEW RETURNED	29-Sep-2020

GENERAL COMMENTS	The prevalence and condition of sarcopenia varies by country, region, and race. It makes sense to clarify and consider this. Inclusion and exclusion criteria Reference 41) of the submitted paper reports “Sarcopenia appears to be highly prevalent in old nursing home residents and hospitalized patients in comparison to community-dwelling individuals”. This paper also includes data from nursing homes. Isn't this one of the reasons for the high prevalence of sarcopenia? I think Fig2 and Fig5 have the same data. However, the total prevalence of sarcopenia is different. Why is the total prevalence of sarcopenia different between Fig2 and Fig5? The same applies to Figures 3 and 6, Figures 4 and 7. Ln.125: What is the definition of "sevear sarcopenia and obesity-related sarcopenia"? How did you find out about these? Ln.157-158: What is the rationale for this idea? Do you have any reference? Ln.160-161: I don't understand the meaning. There is little consideration of the results. For example, why is the prevalence of sarcopenia high in China? Why is the prevalence of sarcopenia in men so high? Why does the prevalence of sarcopenia depend on diagnostic criteria? Why does the prevalence of sarcopenia vary depending on how muscle mass is measured?

REVIEWER	Marcus Borges Universidade de Sao Paulo Hospital das Clinicas, Psychiatry
REVIEW RETURNED	03-Nov-2020

GENERAL COMMENTS	I would like to thank the authors for the submission of their manuscript entitled: "Prevalence of Sarcopenia among the Elderly in China: A Systematic Review and Meta-analysis". This study aimed to estimate the overall prevalence of sarcopenia among the elderly in China. Clear strengths of the study are: 1) The authors conducted a systematic review and meta-analysis which was based on population-based studies that investigated the prevalence of sarcopenia among Chinese older people. 2) This study seems to be relevant within a field of research (e.g. on Geriatrics or Endocrinology);

	3) The subject addressed is worthy of investigation; 4) This study is interesting and novel. In addition, the information presented was new; 5) The authors concluded that: "This systematic review is the first estimation of the pooled prevalence of sarcopenia among the elderly in China. The results suggest that China may have the largest sarcopenia patient count. Therefore, more attention should be given to the prevention and control of sarcopenia." The manuscript is not well written with some grammar errors which requires a language revision by a professional copywriter. I cannot recommend publication of the manuscript. Below my main comments. 1) Title: It identifies properly the study as both systematic review and meta-analysis. 2) Abstract: It provides a structured summary including key messages. However, this section doesn't seem to be accurate, balanced and complete. On page 3, line 27: "elderly Chinese people" or "Chinese older people"? Please, check the term used. On page 3, lines 27-28: The authors reported that several studies have investigated: "the prevalence of sarcopenia among elderly Chinese people" in a repetitive way (it had already been written in subheading – Objectives, as the authors can note). Please, check if this sentence is really necessary or delete it in order to avoid redundancy. On page 3, lines 30-32: Study eligibility criteria and synthesis methods are missing. Besides, data sources were described in subheading: "Setting"? In the body of the text, "setting" was related to "urban" versus "rural" areas. Please, clarify it. On page 3, line 35 (subheading – Outcomes): the outcomes are not clearly defined: "The sex-specific prevalence of sarcopenia at 95% confidence interval (CI) was also calculated". The authors did not mention "3 different criteria used for the definition of sarcopenia." Could you explain it? On page 3, line 40-41(subheading – Conclusion): The authors concluded that: "This systematic review is the first estimation of the pooled prevalence of sarcopenia among the elderly in China." in a repetitive way (it had already been written in subheading – Strengths, as the authors can note). For the sake of a better understanding, check if this sentence is really necessary or delete it in order to avoid redundancy, please. On page 3, line 49 (subheading – Strengths and limitations): Why did not the authors use a cut-off (e.g. "> 60 years old") as an inclusion criterion in order to avoid the major limitation of this study: "Not all studies presented data according to age group"? On page 3, line 50 (subheading – Strengths and limitations): systematic review registration number? 3) Introduction section: The research question or study objective is not clearly defined. On page 4, lines 57-65: The authors described 3 different criteria for

	the definition of sarcopenia and reported that “2 diagnostic criteria recommended are similar”. Could the authors explain the reason for this statement, please? It is unclear if the authors would have compared 3 or 2 criteria regarding the prevalence of sarcopenia, so clarify it. On page 4, line 65: Why did the authors use the word: “cutting” instead of “cut-off”? Could the authors explain the term used, please? On page 4, line 69-71: The authors reported that the objectives of this study are: “to systematically reveal the prevalence of sarcopenia in elderly Chinese and to provide a scientific basis for the work of clinical departments and concerned government agencies.” However, for the sake of a better understanding, review the aim of this study (in Introduction section) in order to keep consistency with the “Objectives” (as mentioned in Abstract section). 4) Methods section: The methods are described sufficiently to allow the study to be repeated. However, On page 4, first paragraph: The authors did not specify “Protocol and registration” and “Research ethics” as it is mentioned and presented in the Supplementary data. Why? Methods section (Subsection – search strategy): On page 4, lines 77-79: It is unclear for the readers how the authors carried out the search strategy using the “search terms” (that were identified according the following databases: PubMed, Web of Science, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), China National Knowledge Infrastructure (CNKI), VIP, China Biology Medicine (CBM), and Wanfang). Could it be repeated the same “search terms” for each database included in the systematic review? Please, explain it for the sake of a better understanding. On page 5, lines 82-83: The authors described that: “The completed Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist is presented in the Supplementary data.” However, it is unclear for the readers what it has exactly been the research question in this study. Please, provide an explicit statement of research question which should be addressed with reference to PICOS (according to PRISMA checklist). Methods section (Subsection – Inclusion and Exclusion Criteria): On page 5, lines 86: The authors reported "(2) …elderly Chinese participants;" It is unclear whether the authors considered “age (e.g. ≥ 60 y.)” as an inclusion criterion. On page 5, lines 87: The authors used the term: “sample size” in a repetitive way (it had already been written in: “(1) sample size of ≥ 100…” , as the authors can note). Please, check if this term is really necessary or delete it in order to avoid redundancy. On page 5, lines 92: The authors used the terms: “subjects were specific population… patients with disability” as exclusion criteria. However, the authors included studies with older people who were living in “nursing home” as the authors can note (in Table 1, “setting”). It is quite confusing for the readers. Please, explain why the authors decided including these participants and setting.

	On page 5, lines 94-98: Please, check and revise the subheading: "Data extraction and risk of bias". Could the authors describe as "Methodological Quality Assessment" instead of "risk of bias"? It was mentioned that: "two independent reviewers… assessed the risk of bias in the included studies using the Newcastle-Ottawa scale (NOS)" instead of "Newcastle-Ottawa Scale (NOS) for assessing the quality of included studies in meta-analysis [reference 13]". That is a quite different statement from the first one... In addition, in the subsection "Statistical analysis" the authors cited that: "Publication bias was evaluated using Begg's test". It is confusing for the readers. Please, specify the risk of bias "within studies" versus "across studies". 5) Results section: The results addressed the research question, but they are not presented clearly. On page 6, lines 117: Please, check the Figure 1 regarding the sentence: "…articles were included based on the criteria shown in Figure 1". As the authors and readers can see, there is no a standard flowchart (e.g. following the PRISMA guideline, it is important to emphasize: "Records screened and excluded by title and abstract"; "Full-text articles excluded, with reasons"; "Full-text articles assessed for elegibility"). Therefore, the authors should correct it, please. On page 6, line 119: Please, check and review the subheadings used in Table 1 (row 1, columns 1, 3, 6, 7): "Study…participate…Method…Diagnose…" respectively. On page 6, line 119: Please, check and review the sentence: "…selected studies' quality was acceptable…" As the authors have previously mentioned in Methods section: "NOS scores range from zero to nine stars." Please, it is unclear for the readers what the results regarding NOS scores and the cut-off to classify "selected studies' quality as acceptable". On page 6, lines 121-133: The authors describe the main results of the meta-analysis. However, it is unclear for the readers, if the authors would have focused on the prevalence of sarcopenia estimated using the AWGS standard (as it was mentioned in Abstract section). Would the authors have compared these findings to IWGS and EWGSOP standards using 2 different measurements (BIA and DEXA)? Please, explain it. On page 6, lines 138-140: Please, check and review the results regarding the potential publication bias. It is unclear for the readers, the p-values ($P > 0.05$) related to Begg's test that suggested "that there were no publication bias". Please, explain it. 6) Discussion section: The discussion and conclusions are not completely justified by the results. In general, the authors should discuss the main findings and compare the results following the different findings (according to 3 standards used to define sarcopenia). It seems to be that the authors focused on the AWGS standard and subgroup analyses. Is it correct? On page 7, line 146: Why did the authors use the word: "elderly syndrome" instead of "geriatric syndrome"? Could the authors explain the term used, please?

	On page 8, line 169: The authors described that: “Second, not all studies presented data according to age group…” How could the authors explain this major limitation of the meta-analysis? 7) Conclusion section: On page 8, line 177: Why did the authors report “the total prevalence of sarcopenia among the elderly in China” of “14.8%” “according to AWGS” in this section? As a main finding, should it be reported in Results section instead of in Conclusion section? On page 8, line 179: Why did the authors use the word: “control” instead of “management”? Could the authors explain the term used, please? The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Comments to the Author

Thank you for this invitation. This is a systematic review addressing the prevalence of sarcopenia in Chinese older adults. It has some merits; however, some important issues need to be addressed.

1. Most of the included studies were conducted in Chinese community-dwelling older adults. However, some of them were conducted in nursing homes or outpatient clinics. It is well-known that the prevalence of sarcopenia is significantly higher in nursing homes than in communities. Therefore, it is not a good idea to merge the data from different settings. Please consider only include the community-dwelling older adults or report the prevalence of sarcopenia in these settings separately. For the same reason, the expression of “the overall prevalence of sarcopenia among the elderly in China” in the abstract was inappropriate.

Answer: Thanks for the comments. The included studies were conducted in nursing homes or outpatient clinics was inappropriate, so we have modified the whole manuscript. We removed these studies and performed the analysis again. Now, all included studies were conducted in Chinese community-dwelling older adults. We think it is appropriate.

2. In the abstract, the authors stated that “the population-based studies that reported…” were selected. I fail to find a similar statement in the methods section. How did the authors know the study populations in the included studies were “population-based”? Generally, the studies conducted in outpatient clinics and nursing homes are not “population-based”.

Answer: We have modified the words and deleted the “population-based”. Also, the meta-analysis only included the community-dwelling older adults’ study.

3. The authors reported the search terms they used in the “search strategy” section. But this is not enough. They should also report the full search strategy for each database to improve the repeatability of the search process. Maybe a supplementary table would be added. Additionally, it is not enough to identify Chinese people using only “China” as the search term.

Answer: We report the full search strategy in supplementary file and modified the words.

4. One inclusion criterion was “a cross-section study or retrospective study”. Were prospective studies excluded? At least, the baseline data of prospective studies are supposed to be valuable to the topic.

Answer: Actually, prospective studies were not excluded in our meta-analysis. We have modified the words and added it in the manuscript.

5. The full names of the databases -“VIP” and “Wanfang” should be given.

Answer: We have modified the words in the manuscript.

6. English writing needs to be improved. The expression of “the elderly” is a bit discriminatory, “older adults” would be better. Besides, the current running title “Prevalence of sarcopenia among” is inappropriate.

Answer: We have modified the words in the manuscript.

7. There are some typos. For example, “AWSG” and “IWSG” in Table 2.

Answer: We apologized for the mistake and have modified the words in the manuscript.

Reviewer: 2

Comments to the Author

The authors evaluated the incidence of sarcopenia in China using meta-analysis. The idea of this study is novel and interesting. I think the manuscript is enough to be published.

There is only one minor issue.

In figure 2, the incidence of sarcopenia is 0.17 (mean), but in the body it is 0.18. The correction is needed.

Answer: We apologized for the mistake and have modified the words in the manuscript. Besides, we performed the analysis again.

Reviewer: 3

Comments to the Author

The prevalence and condition of sarcopenia varies by country, region, and race. It makes sense to clarify and consider this.

Inclusion and exclusion criteria

Reference 41) of the submitted paper reports “Sarcopenia appears to be highly prevalent in old nursing home residents and hospitalized patients in comparison to community-dwelling individuals”. This paper also includes data from nursing homes. Isn't this one of the reasons for the high prevalence of sarcopenia?

Answer: Thanks for the comments. The included studies were conducted in nursing homes or outpatient clinics was inappropriate, so we have modified the whole manuscript. We removed these studies and perform the analysis again. Now, all included studies were conducted in Chinese community-dwelling older adults. We think it is appropriate.

I think Fig2 and Fig5 have the same data. However, the total prevalence of sarcopenia is different. Why is the total prevalence of sarcopenia different between Fig2 and Fig5? The same applies to Figures 3 and 6, Figures 4 and 7.

Answer: We admire the reviewer`s carefulness and earnest. Because some studies only provided the overall prevalence of sarcopenia, no details of females and males, the data of Fig2 and Fig5 were not the same. The number of included studies in Fig2 was more than Fig5, so the results were different. The same applies to Figures 3 and 6, Figures 4 and 7. We think the difference was little influence on our results.

Ln.125: What is the definition of "sevear sarcopenia and obesity-related sarcopenia"? How did you find out about these?

Answer: The data of “severe sarcopenia and obesity-related sarcopenia” were provided by included studies. While, because we removed studies conducted in nursing homes and re-analysis, the studies about severe sarcopenia and obesity-related sarcopenia were few. They were not suitable for subgroup analysis (Studies less than 3). So, we also removed the data about "severe sarcopenia and obesity-related sarcopenia" in the manuscript.

Ln.157-158: What is the rationale for this idea? Do you have any reference?

Answer: The prevalence of sarcopenia among females was higher than that in males in other meta-analysis. The more protein intake and exercise daily will increase the muscle of the body. We think it may be the reason of low prevalence of sarcopenia in males.

Ln.160-161: I don't understand the meaning.

Answer: It's the principle of BIA methods. It may be a little obscure. And we agree to delete the words if it not necessary.

There is little consideration of the results. For example, why is the prevalence of sarcopenia high in China? Why is the prevalence of sarcopenia in men so high? Why does the prevalence of sarcopenia depend on diagnostic criteria? Why does the prevalence of sarcopenia vary depending on how muscle mass is measured?

Answer: We have modified the words in Discussion and added some consideration of the results. Thanks for the comments.

Reviewer: 4

Comments to the Author

I would like to thank the authors for the submission of their manuscript entitled: "Prevalence of Sarcopenia among the Elderly in China: A Systematic Review and Meta-analysis". This study aimed to estimate the overall prevalence of sarcopenia among the elderly in China. Clear strengths of the study are:

1) The authors conducted a systematic review and meta-analysis which was based on population-based studies that investigated the prevalence of sarcopenia among Chinese older people.

2) This study seems to be relevant within a field of research (e.g. on Geriatrics or Endocrinology);

3) The subject addressed is worthy of investigation;

4) This study is interesting and novel. In addition, the information presented was new;

5) The authors concluded that: "This systematic review is the first estimation of the pooled prevalence of sarcopenia among the elderly in China. The results suggest that China may have the largest sarcopenia patient count. Therefore, more attention should be given to the prevention and control of sarcopenia."

The manuscript is not well written with some grammar errors which requires a language revision by a professional copywriter. I cannot recommend publication of the manuscript.

Below my main comments.

1) Title: It identifies properly the study as both systematic review and meta-analysis.

2) Abstract: It provides a structured summary including key messages. However, this section doesn't seem to be accurate, balanced and complete.

On page 3, line 27: "elderly Chinese people" or "Chinese older people"? Please, check the term used.

Answer: Thanks for the advice. We have modified the words to "Chinese older people" in the manuscript.

On page 3, lines 27-28: The authors reported that several studies have investigated: "the prevalence of sarcopenia among elderly Chinese people" in a repetitive way (it had already been written in subheading – Objectives, as the authors can note). Please, check if this sentence is really necessary or delete it in order to avoid redundancy.

Answer: We have rewritten the Abstract.

On page 3, lines 30-32: Study eligibility criteria and synthesis methods are missing. Besides, data sources were described in subheading: “Setting”? In the body of the text, “setting” was related to “urban” versus “rural” areas. Please, clarify it.

Answer: We have modified the words.

On page 3, line 35 (subheading – Outcomes): the outcomes are not clearly defined: “The sex-specific prevalence of sarcopenia at 95% confidence interval (CI) was also calculated”. The authors did not mention “3 different criteria used for the definition of sarcopenia.” Could you explain it?

Answer: We have modified the words.

On page 3, line 40-41(subheading – Conclusion): The authors concluded that: “This systematic review is the first estimation of the pooled prevalence of sarcopenia among the elderly in China." in a repetitive way (it had already been written in subheading – Strengths, as the authors can note). For the sake of a better understanding, check if this sentence is really necessary or delete it in order to avoid redundancy, please.

Answer: Thanks for the comments. “Strengths” was journal requirement in submission . Sometimes, “Strengths” was only provided to reviewers, in order to make the reviewers get the point easily, not showed in the publication. So, we think it is necessary. While if “Strengths” will be published in the paper, we are glad to delete it in order to avoid redundancy.

On page 3, line 49 (subheading – Strengths and limitations): Why did not the authors use a cut-off (e.g. “> 60 years old”) as an inclusion criterion in order to avoid the major limitation of this study: “Not all studies presented data according to age group”?

Answer: Actually, in our meta-analysis, we used a cut-off (e.g. “> 60 years old”) as an inclusion criterion. The prevalence of sarcopenia might be different in 60–70- and 80–90-years old people. Some studies provided the details of 60–70, 70–80, and >80 years old people, while some studies only provided the overall prevalence of sarcopenia of the > 60 years old people. So, we didn`t perform the meta-analysis in subgroup of different age-bracket. Age distribution in different studies may influence on our results.

On page 3, line 50 (subheading – Strengths and limitations): systematic review registration number?

Answer: We have registered the systematic review in PROSPERO. Due to the COVID-19, the registration of PROSPERO team was delayed. We have contacted with PROSPERO team, and they told us they will finish the registration as soon as possible. We will also update the information of systematic review registration number in the future.

3) Introduction section: The research question or study objective is not clearly defined.

On page 4, lines 57-65: The authors described 3 different criteria for the definition of sarcopenia and reported that “2 diagnostic criteria recommended are similar”. Could the authors explain the reason for this statement, please? It is unclear if the authors would have compared 3 or 2 criteria regarding the prevalence of sarcopenia, so clarify it.

Answer: Several different clinical diagnostic criteria for sarcopenia have been reported worldwide. Among these, the 2010 European Working Group on Sarcopenia in Older People (EWGSOP) consensus diagnostic criteria have been most widely adopted and used in many cohort studies involving community-dwelling older adults. Based on the EWGSOP criteria, slow walking speed (0.8 m/sec) or low grip strength (<30 kg for men and <20 kg for women) are indications for muscle mass measurement; sarcopenia is diagnosed if muscle mass is reduced based on sex-specific criteria. The AWGS criteria are consistent with those of the EWGSOP; however, the cut-off values for muscle strength (<26 kg for men and <18 kg for women), walking speed (0.8 m/sec), and muscle mass (7.0 kg/m2 for men and 5.4 kg/m2 for women measured using dual X-ray absorptiometry and 7.0 kg/m2 for men and 5.7 kg/m2 for women measured using bioimpedance analysis) have been revised according to data from regional cohort studies among Asian populations.

On page 4, line 65: Why did the authors use the word: “cutting” instead of “cut-off”? Could the authors explain the term used, please?

Answer: We have modified the words.

On page 4, line 69-71: The authors reported that the objectives of this study are: “to systematically reveal the prevalence of sarcopenia in elderly Chinese and to provide a scientific basis for the work of clinical departments and concerned government agencies.”

However, for the sake of a better understanding, review the aim of this study (in Introduction section) in order to keep consistency with the “Objectives” (as mentioned in Abstract section).

Answer: We have modified the words.

4) Methods section: The methods are described sufficiently to allow the study to be repeated. However, On page 4, first paragraph: The authors did not specify “Protocol and registration” and “Research ethics” as it is mentioned and presented in the Supplementary data. Why?

Answer: Because they were the requirement of the Journal.

Methods section (Subsection – search strategy): On page 4, lines 77-79: It is unclear for the readers how the authors carried out the search strategy using the “search terms” (that were identified according the following databases: PubMed, Web of Science, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), China National Knowledge Infrastructure (CNKI), VIP, China Biology Medicine (CBM), and Wanfang). Could it be repeated the same “search terms” for each database included in the systematic review? Please, explain it for the sake of a better understanding.

Answer: We report the full search strategy in supplementary file and modified the words.

On page 5, lines 82-83: The authors described that: “The completed Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist is presented in the Supplementary data.” However, it is unclear for the readers what it has exactly been the research question in this study. Please, provide an explicit statement of research question which should be addressed with reference to PICOS (according to PRISMA checklist).

Answer: We have added the words in the manuscript.

Methods section (Subsection – Inclusion and Exclusion Criteria): On page 5, lines 86: The authors reported "(2) …elderly Chinese participants;" It is unclear whether the authors considered “age (e.g. ≥ 60 y.)” as an inclusion criterion.

Answer: Actually, in our meta-analysis, we used a cut-off (e.g. “> 60 years old”) as an inclusion criterion.

On page 5, lines 87: The authors used the term: “sample size” in a repetitive way (it had already been written in: “(1) sample size of ≥ 100…” , as the authors can note). Please, check if this term is really necessary or delete it in order to avoid redundancy.

Answer: We have modified the words.

On page 5, lines 92: The authors used the terms: “subjects were specific population… patients with disability” as exclusion criteria. However, the authors included studies with older people who were living in “nursing home” as the authors can note (in Table 1, “setting”). It is quite confusing for the readers. Please, explain why the authors decided including these participants and setting.

Answer: Thanks for the comments. The included studies were conducted in nursing homes or outpatient clinics was inappropriate, so we have modified the whole manuscript. We removed these studies and perform the analysis again. Now, all included studies were conducted in Chinese community-dwelling older adults. We think it is appropriate.

On page 5, lines 94-98: Please, check and revise the subheading: “Data extraction and risk of bias”. Could the authors describe as “Methodological Quality Assessment” instead of “risk of bias”? It was mentioned that: “two independent reviewers… assessed the risk of bias in the included studies using the Newcastle-Ottawa scale (NOS)” instead of “Newcastle-Ottawa Scale (NOS) for assessing the quality of included studies in meta-analysis [reference 13]”. That is a quite different statement from the first one... In addition, in the subsection “Statistical analysis” the authors cited that: “Publication bias was evaluated using Begg’s test”. It is confusing for the readers. Please, specify the risk of bias “within studies” versus “across studies”.

Answer: We have modified the words as advice. Thanks for the advice.

5) Results section: The results addressed the research question, but they are not presented clearly.

On page 6, lines 117: Please, check the Figure 1 regarding the sentence: “…articles were included based on the criteria shown in Figure 1”. As the authors and readers can see, there is no a standard flowchart (e.g. following the PRISMA guideline, it is important to emphasize: “Records screened and excluded by title and abstract”; “Full-text articles excluded, with reasons”; “Full-text articles assessed for eligibility”). Therefore, the authors should correct it, please.

Answer: We have modified the words in the Figure 1.

On page 6, line 119: Please, check and review the subheadings used in Table 1 (row 1, columns 1, 3, 6, 7): “Study…participate…Method…Diagnose…” respectively.

Answer: We have modified the words as advice.

On page 6, line 119: Please, check and review the sentence: "…selected studies' quality was acceptable…" As the authors have previously mentioned in Methods section: "NOS scores range from zero to nine stars." Please, it is unclear for the readers what the results regarding NOS scores and the cut-off to classify "selected studies' quality as acceptable".

Answer: We have modified the words.

On page 6, lines 121-133: The authors describe the main results of the meta-analysis. However, it is unclear for the readers, if the authors would have focused on the prevalence of sarcopenia estimated using the AWGS standard (as it was mentioned in Abstract section). Would the authors have compared these findings to IWGS and EWGSOP standards using 2 different measurements (BIA and DEXA)? Please, explain it.

Answer: We think AWGS standard is more suitable to estimate the prevalence of sarcopenia in China. And we also estimated using IWGS and EWGSOP standards. Besides, the subgroup analysis of 2 different measurements (BIA and DEXA) according 3 different standards were also performed.

On page 6, lines 138-140: Please, check and review the results regarding the potential publication bias. It is unclear for the readers, the p-values ($P > 0.05$) related to Begg's test that suggested "that there were no publication bias". Please, explain it.

Answer: The rank correlation test introduced by Begg and Mazumdar is extensively used in meta-analysis to test for publication bias in clinical and epidemiological studies. It is based on correlating the standardized treatment effect with the variance of the treatment effect using Kendall's tau as the measure of association. $P > 0.05$ of Begg's test means there is publication bias. More details of Begg`s test can be found in reference.

Begg CB, Mazumdar M (1994) Operating characteristics of a rank correlation test for publication bias. Biometrics 50 (4): 1088–1101

6) Discussion section: The discussion and conclusions are not completely justified by the results. In general, the authors should discuss the main findings and compare the results following the different findings (according to 3 standards used to define sarcopenia). It seems to be that the authors focused on the AWGS standard and subgroup analyses. Is it correct?

Answer: We think AWGS standard is more suitable to estimate the prevalence of sarcopenia in China. We also performed the subgroup analyses in another two standards.

On page 7, line 146: Why did the authors use the word: “elderly syndrome” instead of “geriatric syndrome”? Could the authors explain the term used, please?

Answer: We have modified the words as advice.

On page 8, line 169: The authors described that: “Second, not all studies presented data according to age group…” How could the authors explain this major limitation of the meta-analysis?

Answer: The prevalence of sarcopenia might be different in 60-70 and 80-90 years old people. Some studies provided the details of 60-70, 70-80, and >80 years old people, while some studies only provided the overall prevalence of sarcopenia of the > 60 years old people. So, we didn`t perform the meta-analysis in subgroup of different age-bracket. Age distribution in different studies might influence on our results, while, we think the effect was very small and the results were acceptable.

7) Conclusion section:

On page 8, line 177: Why did the authors report “the total prevalence of sarcopenia among the elderly in China” of “14.8%” “according to AWGS” in this section? As a main finding, should it be reported in Results section instead of in Conclusion section?

Answer: We have deleted the words in conclusion as advice.

On page 8, line 179: Why did the authors use the word: “control” instead of “management”? Could the authors explain the term used, please?

Answer: We have changed the words as advice.

VERSION 2 – REVIEW

REVIEWER	Ming Yang Sichuan University, The Center of Gerontology and Geriatrics
REVIEW RETURNED	12-Dec-2020

GENERAL COMMENTS	I would like to thank the authors for their effort to improve this manuscript. My comments have been addressed and revised. Just some minor issues need to be considered from my side. 1. The authors searched both PubMed and Medline. Why? PubMed includes Medline database. There are more than twenty entrances to medline database, of which, PubMed is the most widely used. Of course, you can search medline via different resources. However, please at least show us where did you search medline in File S1.

	For example, Medline via EBSCO or Medline via OvidSP, etc. 2. There is a typo in the search strategy for PubMed, please remove the " before muscle mass. 3. The authors did not use MESH words in their search strategy. Why ? This might induce the omitting of potentially eligible papers.

REVIEWER	S Takahashi National Hospital Organisation Hamada Medical Center, Surgery
REVIEW RETURNED	05-Jan-2021

GENERAL COMMENTS	It is interesting that the prevalence of sarcopenia in China is high. Lin 152-155: What is the rationale for this? Please indicate references Lin 157-158: What is the rationale for this? Please indicate references Lin165: It is described as “Sarcopenia is characterized by deterioration of muscle mass and function that occurs with aging” at the beginning of “Introduction” . Isn't it natural that the prevalence increases with aging? Lin167-168: What are the definitions of Urban and Rural?

VERSION 2 – AUTHOR RESPONSE

Response to Reviewer: 1

1. The authors searched both PubMed and Medline. Why? PubMed includes Medline database. There are more than twenty entrances to medline database, of which, PubMed is the most widely used. Of course, you can search medline via different resources. However, please at least show us where did you search medline in File S1. For example, Medline via EBSCO or Medline via OvidSP, etc.

Answer：Thanks for the good comments. PubMed usually includes Medline database, for avoiding potential missing reference, so we also searched Medline. After our careful consideration, we decide to adopt your advice, only using PubMed. We modified the related words and figure in the manuscript. We hope the manuscript is better now.

2. There is a typo in the search strategy for PubMed, please remove the " before muscle mass.

Answer：Thanks for the comments, and we have removed the " before muscle mass.

3. The authors did not use MESH words in their search strategy. Why ? This might induce the omitting of potentially eligible papers.

Answer：We did not use MESH words in search strategy, because we used PubMed, Web of Science, CNKI, CQVIP, and Wanfang databases. MESH words strategy may be not suitable to be used in Chinese databases (CNKI, CQVIP, and Wanfang). So, we think that our search strategy (search terms in abstract) can obtain more comprehensive literature, avoiding to omit of potentially eligible papers.

Response to Reviewer: 3

Lin 152-155: What is the rationale for this? Please indicate references

Answer：We add the reference [42] for the support the reason.

Lin 157-158: What is the rationale for this? Please indicate references

Answer：I am sorry that this part was not clear in the original manuscript. It may be the words that cause the misunderstanding. Men eat and exercise more than women and we think there is no need to add reference. For better understanding, we also revised the words in the manuscript (Line166-167).

Lin165: It is described as “Sarcopenia is characterized by deterioration of muscle mass and function that occurs with aging” at the beginning of “Introduction” . Isn't it natural that the prevalence increases with aging?

Answer：Thank you for your advice. The prevalence increases with aging, not belong to our finding, so we also modified the words (Line174).

Lin167-168: What are the definitions of Urban and Rural?

Answer：Cities are also called urban settlements, which generally include residential areas, industrial and commercial areas, and have the function of administrative jurisdiction. The administrative jurisdiction function of a city may involve more areas than its own, including residential areas, streets, hospitals, schools, public green space, office buildings, commercial stores, squares, parks, and other public facilities.

Rural areas have specific natural landscape and socio-economic conditions, which refers to the place where workers mainly engaged in agricultural production live in, which is different from cities and

towns where people engaged in agriculture live in. Rural area refers to the agricultural area, including market towns and villages. It is dominated by agricultural industry (natural economy and primary industry), including various farms (including livestock and aquaculture farms), forest farms, horticulture, and vegetable production.

The main differences between rural and urban areas are based on official administrative planning, natural location, and the level of economic development.